

Crowdabunga! The Crowd Challenges of TMNT: Mutant Mayhem

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ABSTRACT

From the original comic book to the TV shows, the universe of the Ninja Turtles is very well-known for its unique characters, whether lead or secondary. To deliver believable and directable crowds for “*Teenage Mutant Ninja Turtles: Mutant Mayhem*” (TMNT: MM), the teams at Mikros Animation had to think of new approaches for the Crowd department to populate dozens of shots while matching the animation and visual style of the show. We faced multiple challenges: the crowd population were close to the camera and had to be diverse while matching the asymmetrical / clay-like style and capabilities of the main characters; the main action takes place in NYC and portrays well known crowded situations; the results had to remain faithful to the 2D teenage style of the show.

In this paper, we detail the tools we developed to tackle these challenges. First, a Character Building Tool was developed to generate hundreds of stylized morphologies from an input of 4 Character Templates, hero garments and pencil strokes. Then, crowd artists took advantage of casting tools to populate their shots, assign them with predefined animations and enhance the final results with procedural behaviors. Finally, post simulation tools were implemented to alter animations and lookdev, based on camera distance.

ACM Reference Format:

Benjamin Dupin, Gaëtan Luzy, and Nicolas Chaverou. 2024. Crowdabunga! The Crowd Challenges of TMNT: Mutant Mayhem. In *Special Interest Group on Computer Graphics and Interactive Techniques Conference Talks (SIGGRAPH Talks '24)*, July 27–August 01, 2024, Denver, CO, USA. ACM, New York, NY, USA, 2 pages. <https://doi.org/10.1145/3641233.3664316>

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SIGGRAPH Talks '24, July 27–August 01, 2024, Denver, CO, USA

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ACM ISBN 979-8-4007-0515-1/24/07

<https://doi.org/10.1145/3641233.3664316>

1 CONTEXT

Mikros Animation is a CGI animation studio dedicated to feature films production based in Paris, Montreal and Bangalore. Its pipeline relies on Maya as a backbone, Houdini for asset building, Golaem for crowds, Katana for assembly and lighting and Arnold for rendering. For TMNT: MM, the team delivered 300 character variations from 4 templates, 70 shots in 9 months with a maximum of 6 artists.

2 PREVIOUS WORKS

For *SpongeBob the Movie*, [Bobant et al. 2021] presented a tool which converts any character rig (made of multiple hierarchies of bones and complex facial deformations) into a crowd-friendly character. For TMNT: MM, we leveraged this tool and updated it to tackle the new challenges we faced, especially body types support. [Dirksen et al. 2012] introduced Multidimensional Rigs to deal with body types and facial animation transfer. While this approach provided great results, the asset creation was manual for each body type and required a fair amount of planning between the departments.

3 ASSETS CONVERSION

3.1 Morphology and Diversity

To allow a wide range of character morphologies, we started with 4 Character Templates (CT): unisex child, female, male and large male. Those body types were modeled and skinned in t-pose and shared the same topology [de Goes and Martinez 2019]. We wrapped every garment available for the show using a custom Houdini SOP tool. Since all bodies share the same topology, the wrapper analyzes the garment and its corresponding body geometry and reproduces the same offset on each CT. It performed well in 90% of the cases and sometimes required local resculpt. Then, skinning was performed in Maya, using a custom script to copy skinning information from the source garments. In addition, we also generated multiple asymmetrical face shapes for each CT using the same approach than body



Figure 1: Variations generated from the Female Template.

types but keeping the underlying facial structure so the facial deformations would work without having to tackle the combinatorial complexity. As mesh connectivity and face features were the same from one sculpt to another, we could easily store them as blend shapes. Finally, CT could be enhanced with nonlinear per-joint ranged scales, to handle size variation as pictured in Fig.1.

3.2 Facial Performance

For facial performances, we relied on the existing workflow which converts any deformations from the hero facial rig to the crowd mesh as blend shapes. We updated it by providing a way of saving facial expressions as templates to be re-used automatically on other templates. In the end, each CT ended up with 100+ facial expression blend shapes, each corresponding to a specific facial rig controller.

3.3 Handling Pencil Strokes

Pencil strokes are an important part of the look of the movie and had to be replicated on the crowds. Strokes were created by the modeling department on each hero body, garment and props using Maya curves. Animators or environment artists could also add curves per shot to refine expressions or speed lines. For the crowds, we only kept the curves defined by the modeling department. For each garment of each CT, we used the same wrapping deformation on the curves than their underlying geometry (cf 3.1). Each curve control value (CV) was also assigned skinning weights to deform properly when characters were animated. We took advantage of the Golaem Fur system to handle those: it stores barycentric coordinates for each hair, CV data and skinning weights when available. Then, at render time, it computes the deformation of each hair according to the parent geometry and generates an Arnold Curve primitive.

4 CROWD SIMULATION AND LAYOUT

4.1 Initial Setup

To set up shots, artists took advantage the Casting tool - it provides a simple UI with vignettes representing every single CT variation validated by the production and instance them in the scene. The casting freezes mesh variation and skeleton overrides of a CT. Shading diversity can be altered to match with the shot color palette. When shot consistency is required, artists uses the Recast tool - it reads a validated shot cache and reverts a brand new simulation from it. It makes sure initial positions / orientations are respected and, more importantly, that the casting is exactly the same.

4.2 Animation Direction

Interestingly enough, on this show, some of the crowd artists were also animators. Thus, with the focus of bringing more life and uniqueness to the shots, they often committed to animate specific



Figure 2: Example of pencil strokes applied to a CT.

actions and use them as crowd vignettes. Then, they would add procedural behaviors to randomize facial expressions or direct gaze / hand IK positions. Finally, they took advantage of post simulation rig inversion tools to fix postures locally or attach props.

5 POST SIMULATION TOOLS

5.1 Animation Steps

The film was mostly animated on 2s, meaning that every pose is held for an extra frame (or more). However, as we needed to perform accurate ground adaptation and motion transitions, the characters were provided motion clips on 1s and simulated at every frame. Then, per shot, artists used the Stepped Tool to alter simulation results - they could layer characters based on camera distance and assign them with different frame rates. When the camera rotated, we filtered the character root out of the process to avoid strobing.

5.2 Rendering Level of Detail

The rendering department built a specific shader for the pencil strokes to maintain their width and texture whatever the distance they were to the camera. As our procedural rendering plugin for the crowds generate curve primitives we could rely on the same process and override properties using Arnold Operators. Following the same workflow than 5.1, the team created layers such as foreground, middleground and background and matched the look of the show by reducing the amount of details for the crowd characters.

6 CONCLUSION

Thanks to the tools we contributed, our team was able to derive more than 300 unique characters from our 4 CT. Those characters included a wide range of facial performances, pencil strokes and could stand seamlessly along key animated hero characters. Our casting tool allowed full control on shot setup and the recast tool handled shot continuity with minimal tedious work. Finally a set of post simulation tools allowed us to blend crowd characters and conform them to the visual style of the show.

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